Vector Quantization of Diffusion Transformers

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Abstract

The goal is to reduce the memory required to store a diffusion transformer's weights while retaining generation quality. We transfer an additive vector-quantization representation to DiT-XL/2: each weight group is approximated by a sum of vectors from several small codebooks. We then examine two calibration choices in sequence. First, five time-specific codebook sets do not improve FID over one shared set at the same total update count: 15.695 versus 15.644, with higher storage. Second, calibrating shared codebooks against the original network's complete output lowers FID from 15.40 to 7.81 at unchanged 4.15 logical bits per DiT parameter. Each comparison generates 10,000 ImageNet256 images with 50 DDPM steps and guidance 1.5; the two comparisons use separate generation cohorts. The findings support full-network calibration in the tested setting, rather than a general claim that diffusion time is unimportant.

1 Introduction and goal

Image generation by a diffusion model repeatedly removes noise using a neural network. DiT replaces the usual convolutional denoiser with a transformer [5]. Storing its many weights requires substantial memory. Weight quantization reduces this requirement by storing a compact approximation of those numerical parameters. Vector quantization approximates groups of weights using entries from small tables, called codebooks, together with indices identifying the selected entries.

Goal. Reduce the logical memory needed to store DiT weights while limiting the loss of generation quality. The study concerns weight storage and quality; it does not measure inference acceleration or training-checkpoint file size.

Main idea and research sequence. Use several small codebooks to represent a weight group as a sum of code vectors, following the additive representation used in VQDM [3]. The motivation is greater approximation flexibility than selecting one vector from one small table. After transferring this representation to DiT, we first test whether separate codebooks for different diffusion times improve quality. The tested time specialization does not improve FID. We then change the calibration objective: after layerwise initialization, optimize codebooks against the output of the entire original network. This second experiment improves FID without increasing storage. These are two calibration studies of the same kind of compact weight representation, not a new distillation method or a fixed-memory single-versus-multiple-codebook ablation.

2 Model and related work

DiT architecture. The public class-conditional DiT-XL/2 checkpoint generates ImageNet images of size 256×256. It operates on noisy VAE latents, divides them into 2×2 patches, and processes the resulting 256 tokens with 28 transformer blocks. Attention and MLP layers transform the tokens;

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time and class embeddings condition the blocks through adaLN-Zero [5]. The final layer predicts noise and a raw learned-variance head. Figure 1 shows the simplified structure.

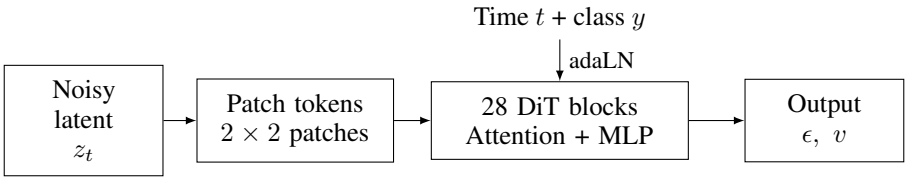


Figure 1: Simplified DiT-XL/2 denoiser, following Peebles and Xie [5]. Positional embeddings, residual connections and final reconstruction details are omitted. Our representation quantizes 140 matrices inside the 28 blocks.

Quantization and calibration. VQDM combines additive vector weight quantization with global output-reconstruction fine-tuning for SDXL and SDXL-Turbo [3]. Here the representation and calibration idea are transferred to DiT. The direct control for the output-calibration experiment is its transferred layerwise stage, rather than the complete original VQDM pipeline. VQ4DiT calibrates codebooks and candidate assignments through transformer-block outputs [1]; it is context, not a reproduced baseline in these tables.

Reconstructing a network's output is a calibration objective. When a frozen full-precision model supplies the target, as here, that objective also provides a knowledge-distillation signal [4]. Quantization describes the compact weight representation; distillation describes the source of the target. Generation quality is measured with the official ADM toolkit [2].

3 Problem statement

Weight representation. Five matrices in each block are quantized: attention QKV and output projections, both MLP projections, and adaLN modulation. There are 140 matrices containing 668,860,416 weights; the complete DiT contains 675,129,632 parameters. Activations, biases and parameters outside these matrices are not quantized.

For a length-d weight subvector, one-codebook quantization would use $\widehat{w} = sC[a]$. The additive representation instead uses

$$\widehat{w}_{\ell,r,g} = s_{\ell,r} \sum_{m=1}^{M} C_{\ell,m}[a_{\ell,r,g,m}], \qquad M = 4,\ K = 256,\ d = 8. \tag{1}$$

Here ℓ is a layer, r an output row, g a subvector, C codebooks, a indices and s row scales. Selecting and summing four vectors provides a more flexible approximation than selecting one from a table of the same size, but uses more index bits. We do not infer a measured fixed-memory advantage over one codebook. The chosen indices require $M \log_2 K/d = 4$ bits per quantized weight. Codebooks, scales and unchanged weights bring the total to 4.150078 logical bits per DiT parameter, excluding the VAE.

First hypothesis: time specialization. Because denoiser inputs change along the diffusion trajectory, a different codebook set for each time window may improve generation relative to a shared set. A *window* here means a consecutive group of generation steps: the 50 DDPM steps are split into five windows of ten steps each. Each window selects its own complete set of additive codebooks. Four codebooks summed within Equation (1) and five time-specific sets of those codebooks are different dimensions of the representation.

Second hypothesis: calibration objective. For the same layerwise-quantized initialization and storage precision, calibrating shared codebook values by reconstructing the entire original network's output lowers FID without increasing logical storage. This direction is tested after the temporal comparison does not show a gain.

Inputs and quality criteria. Let $f_\theta(z_t, t, y) = (\epsilon_\theta, v_\theta)$ denote the original full-precision DiT and $f_{Q(C)}$ the quantized model. The head v is a raw prediction used to parameterize variance, not variance itself. Real-image VAE latents z_0 are noised as

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\, \eta, \qquad \eta \sim \mathcal{N}(0, I), \quad t \sim \mathrm{Uniform}\{0, \ldots, 999\}. \tag{2}$$

The original and quantized networks receive identical inputs. Training times on the original 1,000-value schedule are distinct from positions 1–50 in the generation trajectory.

The primary criterion is lower FID on 10,000 generated images. Storage must be reported alongside FID: temporal specialization can add codebooks, whereas global calibration of their values preserves storage. FID compares Inception-feature means and covariances; sFID, IS, precision and recall are auxiliary criteria. Output MSE is a diagnostic, not a replacement for generation quality. Both hypotheses are examined in fixed configurations without across-training-seed confidence intervals.

4 Solution: properties and algorithm

4.1 Time-specific codebook sets

The layerwise calibration objective approximates each matrix's output,

$$\mathcal{L}_{\mathrm{layer}} = \sum_\ell \|X_\ell W_\ell - X_\ell \widehat{W}_\ell\|_F^2, \tag{3}$$

using calibration activations X_ℓ. The temporal variant gathers separate input statistics for the five windows and refits separate codebook values, retaining common assignments and row scales. The control uses one shared set for 1,000 refit updates; the temporal variant uses five sets with 200 updates each. Thus the total update count is matched. It is not a guarantee that wall-clock cost is identical.

Generation chooses the set associated with the current window; codebook values are fixed during generation. Storing additional sets increases logical bpp. The experiment therefore asks whether this extra specialization produces a quality gain sufficient to motivate its memory cost.

4.2 Full-network output calibration

The second direction returns to one shared set across all times and starts from the same layerwise initialization as its direct control. Instead of optimizing only separate layer outputs, codebooks minimize

$$\mathcal{L}_{\mathrm{rec}} = \mathrm{MSE}(\epsilon_{Q(C)}, \epsilon_\theta) + 0.1\, \mathrm{MSE}(v_{Q(C)}, v_\theta). \tag{4}$$

Only codebook values change. Assignments, row scales, biases and all other weights remain fixed. The target is the original network's prediction, not the sampled noise η. The variance-head term is regression, not a DDPM variational-bound loss. Guidance is not applied during calibration; labels are replaced by the null class with probability 0.1.

Property 1: storage is unchanged by value-only calibration. For matrices of size $o_\ell \times i_\ell$, the shared representation has

$$B_{\mathrm{bits}} = \sum_\ell \frac{o_\ell i_\ell}{d} M \lceil \log_2 K \rceil + 16 \sum_\ell MKd + B_{\mathrm{fixed}}, \tag{5}$$

where indices form the first term, FP16 codebooks the second, and unchanged scales and other parameters form B_{fixed}. Changing codebook values does not change array sizes or precisions, so logical storage is invariant. This property applies to global value-only calibration, not to adding temporal sets.

Property 2: output error and noise-prediction error. For noise outputs $S = \epsilon_{Q(C)}$, $T = \epsilon_\theta$ with D elements, define $R(S) = \mathbb{E}\|S - \eta\|^2/D$ and $D_{\mathrm{rec}} = \mathbb{E}\|S - T\|^2/D$ on a common input distribution. With finite second moments, the L^2 triangle inequality gives

$$\sqrt{R(S)} \le \sqrt{D_{\mathrm{rec}}} + \sqrt{R(T)}, \tag{6}$$

since $S - \eta = (S - T) + (T - \eta)$. Reconstruction controls one term in this bound; it does not guarantee lower actual noise error or FID.

Algorithm for global calibration.

1. Load the layerwise-calibrated model and its original full-precision reference. Freeze all student parameters except codebook master values.
2. For 500 updates, sample batches of 16 training latents and identical noised inputs for both networks using Equation (2).
3. Minimize Equation (4) by backpropagating through the entire quantized denoiser and applying Adam with learning rate 10^{-5} to codebooks only.
4. Monitor a fixed held-out probe; export FP16 codebooks at step 500. Verify unchanged assignments and frozen parameters before generation.
5. Generate matched 10,000-image sets for the original, layerwise-only and globally calibrated models; evaluate all with the same ADM configuration.

FP32 codebook master values are optimized through FP16 casts, with reconstructed weights used in FP32 computation. The implementation's cast gradients are used without claiming exact derivatives of discrete rounding. Gradients are not unrolled through the full sampling trajectory.

5 Experiment

Objectives and temporal comparison. The first experiment compares five window-specific sets with a shared set at the same 1,000-update total. Each variant generates 10,000 images with DDPM50 and CFG 1.5, using the same historical generation cohort within this comparison. The source measurements and setup are retained in the pinned research snapshot. Table 1 reproduces those measurements.

Data for output calibration. The second experiment tests the addition of 500 full-network calibration updates, checks fixed storage and measures auxiliary quality. Initial layerwise calibration used 256 teacher-generated DDPM trajectories. Global calibration uses 5,000 fixed ImageNet training images from two fixed shards, encoded with the MSE VAE: 4,000 latents for updates and 1,000 reserved. A fixed probe uses 96 input/noise pairs at six times from the reserved subset; its MSE is not an average over all 1,000 reserved images. Saved probes at updates 0, 50, 100, 200 and 500 are monitored before FID evaluation. The final500 point also has the lowest saved probe error.

Output-calibration evaluation. All three models generate 10,000 images, ten per ImageNet class, with common sample records and paired initial/per-step noise, selection seed 20261103, **DDPM50**, CFG 1.5, `sd-vae-ft-mse`, FP32, disabled TF32 and mathematical attention. This is a separate generation cohort from the temporal study. Only differences *within* each comparison are interpreted; temporal-study FIDs are not subtracted from output-calibration FIDs.

The official ADM Inception graph and ImageNet256 reference are fixed. FID/sFID use stored full-dataset reference statistics; precision/recall use its 10,000 reference images. Output-calibration archives extend earlier same-protocol 5,000-image prefixes, with all metrics recomputed on the complete 10,000-image sets. No 5,000-image result is compared with a 10,000-image control.

6 Results and error analysis

6.1 Time specialization did not improve FID

Table 1 compares the shared and time-specific codebooks. Five sets yield slightly higher FID despite additional storage. Window-local reconstruction errors decrease by 1.27–3.96%, but this does not translate into an observed generation-quality gain. The result does not support the expected benefit of this specialization; it neither establishes statistical equivalence nor implies that diffusion time is unimportant.

6.2 Full-network calibration improved FID

Tables 2 and 3 compare the original DiT, transferred layerwise representation and that representation after global calibration. FID falls from 15.40 to 7.81, a relative reduction of 49.3%, at unchanged

Table 1: Temporal-codebook experiment: ImageNet256, 10,000 images, DDPM50 and CFG 1.5. Update counts are matched in total. The cohort is separate from Table 2.

Codebook sets	Refit updates	Logical bpp	FID↓
One shared set	1 000	4.1501	15.644
Five time-specific sets	5×200	4.2588	15.695

4.15 bpp. A saved check records zero changes among 334,430,208 assignments. This supports adding output calibration in this configuration.

Table 2: Model–data–criteria summary for output calibration. IN256/10k means ImageNet 256×256, 10,000 generated images per model. All rows share DDPM50, CFG 1.5, MSE VAE and paired noise. Logical bpp excludes the VAE.

Model	Evaluation data	bpp	FID↓	sFID↓
FP reference	IN256 / 10k	32	6.098	21.675
Layerwise VQDM	IN256 / 10k	4.1501	15.402	27.601
VQDM + output calibration	IN256 / 10k	4.1501	7.814	23.367

Table 3: Auxiliary quality criteria on the same ImageNet256/10k sets as Table 2. Higher values are preferred.

Model	IS↑	Precision↑	Recall↑
FP reference	253.827	0.7904	0.6659
Layerwise VQDM	145.312	0.6491	0.7115
VQDM + output calibration	211.549	0.7427	0.6770

Remaining error and trade-offs. sFID, IS and precision improve, but recall decreases from 0.7115 to 0.6770. There is a coverage trade-off, and FID remains 1.72 above the original model. Held-out teacher-noise MSE falls from 0.001162851 to 0.001007803, or 13.33%. This same-input prediction error and generation FID measure different effects and are reported separately.

Measurement and uncertainty. Independent replay with a freshly fetched official ADM CLI reproduces checked FIDs within 10^{-5}, including a published fixed ADM fixture. Some recall values differ at the fourth decimal place. This validates checked FID calculations, not bitwise identity of every metric or reproduction of another paper's compression results.

Neither comparison estimates variation across independently trained quantizers. The output-calibration experiment reuses a fixed generation cohort exploratorily and trains on two fixed ImageNet shards; balanced evaluation differs from natural class frequencies. There is no compute-matched alternative calibration objective, so the observed gain demonstrates the benefit of adding this global stage, not its superiority to every other use of 500 updates. There is also no matched-memory one-versus-four-codebook quality comparison.

7 Conclusion

The work starts from a storage problem: approximate DiT weights compactly using several additive codebooks while retaining generation quality. The tested temporal specialization does not improve FID over shared codebooks, despite using more memory. The subsequent change to full-network output calibration improves FID from 15.40 to 7.81 at unchanged logical storage. A gap to the original model and reduced recall remain. Independent repeats and compute-matched objective comparisons are needed to determine how broadly the result holds.

Artifacts. The practicum repository contains this manuscript, the original one-slide talk and the current six-slide presentation. Code and experiment history remain in the research repository.

References

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